

SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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SpamShield: SMS Spam Detection System

DSA0304-NATURAL LANGUAGE PROCESSING

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Artificial Intelligence and Data Science

The Problem: SMS Spam

Phishing & fraud

- Fake bank alerts and delivery notices ('smishing')
- Links lead to credential-stealing sites
- Prize and lottery scams ask for fees or card details

Wasted attention

- Unsolicited promotions flood inboxes
- Important messages (OTPs, alerts) get buried
- Users lose trust in SMS as a channel

Hard to filter

- Short, informal text with slang and typos
- Spammers tweak wording to dodge keyword lists
- Rules go stale; learned models adapt

Why machine learning?

Instead of hand-written rules, SpamShield learns the statistical patterns of spam from labeled examples, then generalizes to new, unseen messages. Retraining on fresh data keeps it current as spam tactics evolve.

How SpamShield Works



Input

Raw SMS text arrives from user or gateway

Clean

Lowercase, remove punctuation, strip stop words, stem tokens

Vectorize

TF-IDF turns words and n-grams into weighted numeric vectors

Classify

Trained model computes spam probability

Verdict

Label plus confidence; spam is blocked or flagged

Technology stack

Python	NLTK / re	scikit-learn	Flask / FastAPI	Pickle / joblib
Core language	Text preprocessing	Vectorizer and models	Prediction API	Model persistence

Data & Feature Engineering

~5,574	~4,827	~747	80 / 20
Labeled messages	Ham (legit)	Spam	Train / test split

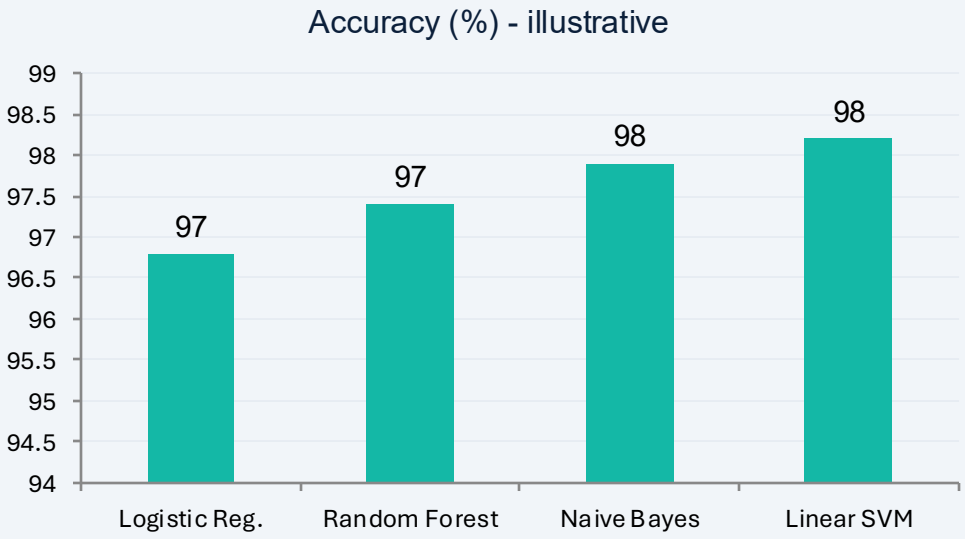
Dataset & preparation

- UCI SMS Spam Collection (English)
- Imbalanced: about 13% spam
- Stratified split preserves class ratio
- Duplicates removed; text normalized
- Cross-validation guards against overfitting

Features

- TF-IDF on unigrams and bigrams
- Message length, word and digit counts
- Presence of URLs, phone numbers, currency symbols
- Capital-letter ratio and exclamation marks
- Urgency words: free, win, claim, urgent

Model Training & Performance



Key insights

- Linear models suit sparse text features
- Naive Bayes: fastest, strong baseline
- SVM: best accuracy, slightly slower
- Tune with grid search on F1-score
- Avoid false positives: never block real texts

98%+	~92%	~95%	<10 ms
Precision	Recall	F1-score	Per-message latency

Metrics are illustrative; replace with your own results.

SpamShield in Action

"WINNER! Claim your free £500 prize now: bit.ly/xyz"

Signals: urgency word, currency symbol, shortened URL

SPAM 98% spam

"URGENT: your account is locked. Verify at secure-bank.co"

Signals: urgency, account threat, suspicious link

SPAM 94% spam

"Running late, see you at the cafe in 10 mins"

Signals: conversational tone, no links or offers

HAM 3% spam

Deployment: wrap the model in a REST endpoint (POST /predict) or embed it in a mobile app or SMS gateway. Sample messages shown for illustration.

Conclusion & Future Work

What we built

- End-to-end SMS spam classifier
- High accuracy with millisecond inference
- Interpretable features and simple retraining
- Easy to deploy as an API or on-device
- Reduces fraud exposure and inbox clutter

What's next

- Transformer models (BERT, DistilBERT)
- Multilingual and code-mixed SMS support
- User feedback loop for continuous learning
- Explainability: highlight why a text was flagged
- Privacy-preserving, on-device inference

Takeaway: classical ML plus good text features delivers a fast, accurate and deployable spam shield.